Research Proposal:

Do Different Deep Learning Models

Explain Distinct Components of the Cortical Neural Code?

האם מודלים שונים של רשתות נוירונים עמוקות מסבירים רכיבים שונים של הקוד העצבי בקליפת המוח?

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1 Research Background

The primate visual system supports rapid object recognition through a distributed hierarchy of cortical areas [Felleman and Van Essen, 1991]. Along the ventral visual pathway, visual information is progressively transformed into representations that support the recognition of objects across substantial variation in their appearance [DiCarlo et al., 2012]. Although the response properties of early visual areas are comparatively well characterized, the computational principles underlying representations in higher-order visual cortex remain incompletely understood [Conwell et al., 2024].

Deep neural networks (DNNs) trained on object-recognition tasks, despite not being optimized to predict neural data, can strongly predict neural responses in the visual cortex through linear mappings from their intermediate activations. This correspondence has established DNNs as promising computational models for investigating higher-order visual processing. A landmark study by Yamins et al. [2014] showed that, within a class of hierarchical convolutional neural networks, models with better object-recognition performance were also more predictive of neural responses in macaque inferior temporal cortex. Importantly, a model optimized only for recognition performance, without using neural data to optimize its internal representations, accurately predicted responses in both IT and V4, with higher model layers corresponding more closely to later stages of the ventral visual hierarchy.

Güçlü and van Gerven [2015] first extended this hierarchical correspondence to the human visual cortex using fMRI, showing that progressively deeper CNN layers better predicted responses in increasingly

higher regions along the ventral visual stream. Eickenberg et al. [2017] later reproduced this spatial gradient across natural-image [Kay et al., 2008] and movie-viewing [Huth et al., 2012] datasets, suggesting that the correspondence between CNN and cortical representational hierarchies generalizes across experimental paradigms. Conwell et al. [2024] systematically compared 224 visual models differing in architecture, training objective, and visual diet. Across controlled comparisons, CNNs and Vision Transformers, as well as vision-only and vision-language models, achieved similar neural predictivity in human occipitotemporal cortex despite their different inductive biases.

2 Research Question, Objectives, and Importance

The scientific background leads to our central research question: Do different models explain distinct aspects of the cortical neural code? More specifically, when models achieve similar overall neural-predictivity scores, to what extent do they capture complementary information about neural responses? Answering this question is important because, if models with different inductive biases provide unique information about neural responses, comparing such models could reveal distinct computational properties of cortical representations. Conversely, if their information is largely redundant, differences in current model architectures and learning objectives may provide limited additional insight into the cortical code, suggesting that a different modeling approach is required. This question therefore relates to the broader debate over whether brain-aligned representations are largely universal across models or retain model-specific inductive biases. Such inducitve biases can arise from differences in architecture, such as CNNs versus Vision Transformers, from learning objectives, such as supervised versus self-supervised learning, or from training with vision alone versus vision-language alignment. For example, comparing a vision-language model such as CLIP with a matched vision-only model can test whether language alignment contributes unique information about cortical responses beyond that captured through visual learning alone.

3 Research Method

One established method for investigating the unique contributions of encoding models is commonality analysis, also known as variance partitioning. This method estimates shared and unique contributions by comparing the variance explained by each predictor separately with that explained by their joint regression model. However, when predictors are correlated, variance partitioning can be affected by suppression effects, such that a predictor that does not predict the target on its own can still be assigned a positive unique contribution [Xiang et al., 2025]. Information theory provides an alternative framework for quantifying the relationship

between encoding-model predictions and neural responses. Mutual information quantifies the amount of information that one variable, or a joint set of variables, provides about a target variable. Partial Information Decomposition (PID), introduced by Williams and Beer [2010], extends mutual information to characterize how information about a target is distributed across multiple sources. For two encoding-model predictions X_1 and X_2, and a neural response Y, PID decomposes their joint mutual information into four components:

$$I(Y; X_1, X_2) = \text{Red}(Y; X_1, X_2) + \text{Unq}_1(Y; X_1 \setminus X_2) + \text{Unq}_2(Y; X_2 \setminus X_1) + \text{Syn}(Y; X_1, X_2). \quad (1)$$

The redundant component quantifies information about the neural response that is shared by both models, whereas each unique component quantifies information provided by one model but not the other. The synergistic component quantifies information available only when the models are considered jointly. Throughout this study, the PID components will be calculated using the BROJA formulation defined by Bertschinger et al. [2014]. For sources X_1, X_2 and target Y, BROJA considers the following definitons:

$$\begin{aligned}
&\Delta(P) = \{Q : \; Q(Y, X_1) = P(Y, X_1), \; Q(Y, X_2) = P(Y, X_2)\}, \\
&I_Q(Y; X_1 \mid X_2) = \text{Unq}_1 + \text{Syn}_Q, \\
&\text{Unq}_1^{\text{BROJA}} = \min_{Q \in \Delta(P)} I_Q(Y; X_1 \mid X_2).
\end{aligned}$$

The set $\Delta(P)$ fixes the two source-target marginals. Under the BROJA assumption that unique information depends only on these marginals, the unique information is fixed across all $Q \in \Delta(P)$, turning its calculation into an optimization problem over the synergistic information. We have derived an analytical solution for Gaussian BROJA PID and are currently preparing a separate manuscript presenting its theoretical proof and empirical evaluation. In our setting, the unique-information components are of particular interest because they quantify whether one encoding model provides information not captured by the other. Importantly, PID can distinguish suppressive contributions from unique information. To illustrate this distinction, consider the Gaussian model:

$$Y \perp N \perp \varepsilon, \qquad X_1 = Y + N + \varepsilon, \qquad X_2 = N,$$

X_2 alone carries no information about Y, and therefore explains no variance in Y Nevertheless, when X_1 and X_2 are considered jointly, X_2 acts as a suppressor variable by removing the nuisance component N from X_1.

The empirical analysis will use neural responses from the Natural Scenes Dataset (NSD), which contains high-resolution 7T fMRI measurements from eight human participants viewing thousands of natural-scene

Commonality Results

n_samples=1000000 | p=50 | seed=42-91 | method=ridge_cv | scale_by_target_variance=False

R^2_X1	R^2_X2	R^2_X12	Unique_X1	Unique_X2	Common	Unexplained
0.5000	-0.0001	0.6666	0.6667	0.1667	-0.1667	0.3334

(a)

PID Method Comparison

n_samples=1000000 | dx1=50 | dx2=50 | dt=50 | seed=42-91 | bias_correction=False

method	Red	Unq1	Unq2	Syn	I(X1;T)	I(X2;T)	I(X1,X2;T)
True Values	0.0000	25.0004	0.0000	14.6239	25.0004	0.0000	39.6243
Analytical BROJA	0.0018	25.0004	0.0000	14.6257	25.0022	0.0018	39.6279

(b)

Figure 1: Comparison between commonality analysis and partial information decomposition in the suppression example. (a) Commonality analysis. (b) Partial information decomposition. (True Values are the values for an unbiased estimator of PID. The second row shows the values of sample from the theoretical covariance). As shown in Figure 1, commonality analysis based on regression can assign a positive unique explained-variance contribution to X_2, despite X_2 carrying no information about Y on its own. In contrast, PID assigns zero unique information to X_2, while its contribution when combined with X_1 is attributed to synergistic information. This agrees with the structure of the example: X_2 is informative only through its interaction with X_1, where it allows the nuisance component N to be removed.

images [Allen et al., 2022]. Each participant viewed approximately 9,000–10,000 distinct images, including a designated set of 1,000 images shared across participants. The participant-specific images will be used to fit the encoding models and select the most predictive DNN layer, while the 1,000 shared images will be reserved as a held-out evaluation set. Using the same held-out stimuli across participants will also allow the resulting information estimates to be compared statistically across participants. For each participant, principal component analysis (PCA) will first be applied to the cortical responses to obtain a lower-dimensional representation of the neural target while retaining most of its variance. For each DNN model, the NSD images will then be passed through the network and activations will be extracted from its intermediate layers. For each layer, ridge regression will be used to map the DNN features onto the PCA-reduced neural responses, and the most predictive layer will be selected according to cross-validated correlation with the neural data. For each pair of DNNs, the predictions of their selected layers on the 1,000 held-out shared images will define the two source variables, X_1 and X_2, while the corresponding PCA-reduced measured cortical responses will define the target Y. The resulting variables will then be modeled as jointly Gaussian, allowing pairwise mutual informations $I(Y; X_1)$ and $I(Y; X_2)$, as well as the joint-source mutual information $I(Y; X_1, X_2)$, to be estimated from their covariance matrices. The four PID components will subsequently be calculated using our closed-form solution for Gaussian BROJA PID. Because finite-sample estimation introduces bias into both mutual-information and PID estimates, Gaussian mutual-information finite sample bias will be corrected using the analytical finite-sample correction for Gaussian differential entropy [Cai et al., 2015], while the Gaussian PID estimates will be corrected for limited-sampling bias following Lorenz et al. [2026]. The analysis will be applied to controlled pairs of models differing along specific sources of inductive bias, including CNNs versus Vision Transformers, supervised versus self-supervised models, and vision-only versus vision-language models. The analysis will reveal all four PID values.

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